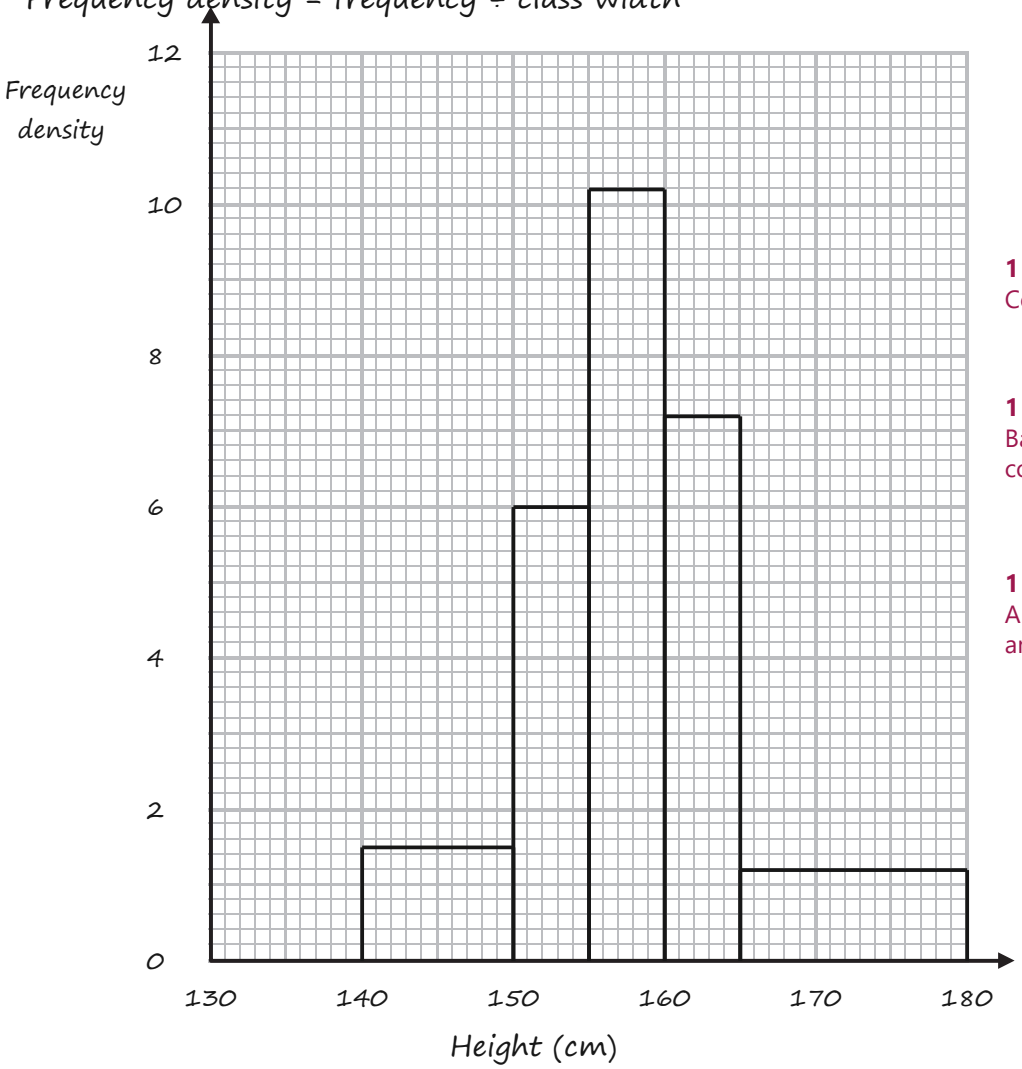
17 The table gives information about the heights of 150 students.

Height (h cm)	Frequency	Width	Density
$140 < h \leqslant 150$	15	10	1.5
$150 < h \leqslant 155$	30	5	6
$155 < h \leqslant 160$	51	5	10.2
$160 < h \leqslant 165$	36	5	7.2
$165 < h \leqslant 180$	18	15	1.2

(a) On the grid, draw a histogram for this information.

Frequency density = frequency ÷ class width



1 mark
Correct densities

1 mark
Bars in the correct proportions

1 mark
All bars, scales and labels correct

(3)

(b) Work out an estimate for the fraction of the students who have a height between 150 cm and 170 cm.

150–165: 30 + 51 + 36 = 117 students.

165–170: (5 ÷ 15) × 18 = 6, so total = 123.

$\frac{123}{150}$

(2)

1 mark
Include one third of the last class

1 mark
Required fraction = 123/150 = 0.82

(Total for Question 17 is 5 marks)